

# Semi-Supervised Multi-View Fusion for Identifying CAP and COVID-19 With Unlabeled CT Images

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**Abstract**—Recently, under the condition of reducing nucleic acid testing for COVID-19 in large population, the computer-aided diagnosis with the chest computed tomography (CT) image has become increasingly important in differential diagnosis of community-acquired pneumonia (CAP) and COVID-19. In practice, there usually exist a mass of unlabeled CT images, especially in regions without adequate medical resources, and the existing diagnosis methods cannot take advantage of the useful information among them. Therefore, it is practical and urgent need to develop a computer-aided diagnosis model that can effectively exploit both labeled and unlabeled samples. To this end, in this paper, we propose a semi-supervised multi-view fusion method for the diagnosis of COVID-19. It explores both the discriminative features from labeled samples and the structure information from unlabeled samples and fuses multi-view features extracted from CT images, including image feature, statistical feature, and lesions specific feature, for improving the diagnostic performance. Specifically, in the proposed model, we utilize semi-supervised learning technique with pairwise constraint regularization to learn the model with both labeled samples and unlabeled samples. Simultaneously, we employ low-rank multi-view constraint to capture latent complementary information among different features from CT images. Experimental results show that the proposed method outperforms the state-of-the-art methods in differential diagnosis of CAP vs. COVID-19.

**Index Terms**—Multi-view fusion, CAP, COVID-19, semi-supervised learning, chest CT image.

## I. INTRODUCTION

CORONAVIRUS disease 2019 (COVID-19) was first identified in December 2019 in Wuhan, China, and has since

spread globally, resulting in an ongoing pandemic [1], [2], [3]. The widely used method of COVID-19 detection is real-time reverse transcription polymerase chain reaction (rRT-PCR). This approach requires a strict laboratory environment and a long time waiting for results. Subject to the normative requirements of sampling, the accuracy of single test of rPT-PCR is hard to meet the actual requirements, often leading to false negative, and multiple repeated tests are often required [4], which is difficult to achieve in the trend of reducing nucleic acid testing for COVID-19 in large population. Fortunately, clinical experience shows that chest CT images can be used as a complementary method for COVID-19 diagnosis [4], [5], [6], [7], [8], [9], [10]. CT images do not require physical transmission and can be used for remote and rapid diagnosis of COVID-19, which makes it particularly suitable for areas with limited medical resources. In addition, CT can also help determine the severity of the disease, which could not be done by rPT-PCR. Therefore, it is urgently necessary to develop an effective CT-based computer-aided diagnosis method for COVID-19.

Because of the superiority of chest CT images based detection of COVID-19, many machine learning diagnosis methods have emerged. For example, Xu et al. proposed a domain adaptation (DA) method to solve the cross-site problem in the diagnosis of COVID-19 based on CT images [11]. Barstugan et al. used manually segmented images of lesion areas from grayscale CT images of the lungs to train the classifier [12]. Wang et al. and Li et al. employed an end-to-end diagnosis approach, which utilizes the chest CT images as input and outputs the classification results with a CNN network [13], [14]. Ozkaya et al. first used CNN to conduct deep feature fusion to the samples and then used the fused features to train a SVM classifier for COVID-19 classification [15]. Shan et al. proposed a neural network model called VB-Net to segment the COVID-19 regions in CT images [16]. Xu et al. analyzed CT images to determine healthy, COVID-19 and other viral cases [17].

Although CT images of many subjects have been accumulated, it is difficult to diagnose all of them manually by the limited number of medical experts. Due to the high transmission rate of COVID-19, the unlabeled samples become much more than the labeled samples in some regions without adequate medical resources and experts. From the perspective of pattern recognition, only part of subjects have diagnosed labels, and the unlabeled subjects are absent in training the diagnosis models in the existing methods [12], [13], [14], [15], [16], [17]. The shortage of labeled CT images seriously restricts the performance of

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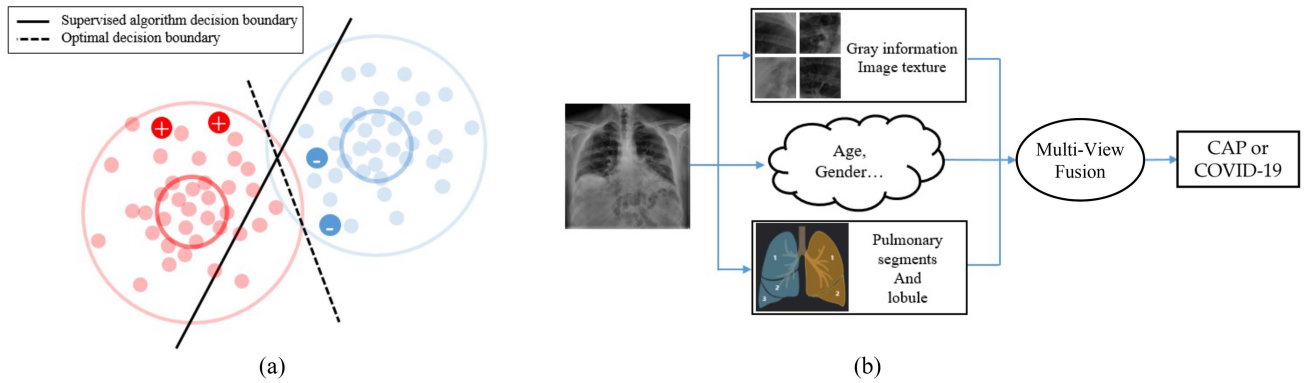


Fig. 1. Two challenges of differential diagnosis of CAP and COVID-19 are simultaneously solved in the proposed framework. (a) Using unlabeled samples to get a better classification/diagnosis boundary. (b) Fusing multi-view features for accurate diagnosis of CAP and COVID-19.

diagnosis algorithms, especially for deep learning based model [18]. For the utilization of unlabeled data, some semi-supervised methods are proposed. For example, Luo et al. proposed an adaptive framework of semi-supervised feature selection for cross-modal data retrieval [19]. Chen et al. developed an effective semi-supervised framework to handle limited labeled data and employed Recurrent Attention Models to extract salient features from unlabeled samples [20]. Yu et al. exploited the correlation between pairs of data by Canonical Correlation Analysis to preserve neighbor relationship of heterogeneous data in the latent common subspace (ASFS) [21]. In addition, the existing COVID-19 diagnosis algorithms mainly focus on the image features of chest CT images, but most of them ignore the manual features, such as lesion position and the number of lesions. There have been a lot of works proving a fact that there exists complementary information between different image features, which can improve the performance of the model. However, most of the previous methods utilize single modal feature [22] or simply concatenated the different features together, which cannot fully capture the latent relationship of data and take advantage of the complementary information among different features from CT images.

To solve these problems, we propose a novel semi-supervised multi-view fusion method for differential diagnosis of community-acquired pneumonia (CAP) and COVID-19, which effectively fuses multi-view features from both labeled and unlabeled subjects for training robust diagnosis model, shown in Fig. 1. A reasonable assumption is that, for CAP and COVID-19 patients, there are some intrinsic correlations among different features from their CT images. For mining the complementary information among different features, we project the data of each view into the feature space and then fuse them for classification. Simultaneously, we introduce the low-rank technique to capture the latent structural relationship of different views of data. To utilize the unlabeled data, we embed domain knowledge into the model by subspace analysis [23], [24], [25]. Domain knowledge can be expressed in many terms, including class labels, pair-wise constraint, and other prior information. We focus on domain knowledge in the form of pair-wise constraints, i.e., pairs of instances known as belonging to the same

class (must-link constraints) or different classes (cannot-link constraints), to learn the structural relationship among labeled and unlabeled subjects. Although the data from different views are heterogeneous, our method is able to effectively fuse them through multi-kernel learning. Most of the previous COVID-19 diagnosis methods are based on supervised learning, and they cannot utilize the information inherently contained in unlabeled data in learning the model. The discriminative features in labeled data and the structure features in unlabeled data are fused in our model.

In summary, this work has the following contributions:

- 1) An effective semi-supervised multi-view fusion method for differential diagnosis of CAP and COVID-19 is proposed. The proposed method effectively fuses multi-view data to diagnose COVID-19, which can make good use of complementary information in different views to improve the accuracy and robustness of the diagnosis model.
- 2) The proposed method can simultaneously learn discriminative features and structure information with both labeled subjects and unlabeled subjects. Due to the large amount of unlabeled data in hospitals and the high cost of manual labeling, compared with supervised models, using semi-supervised learning techniques with features from unlabeled samples is more suitable to solve clinical diagnosis problems, and it is expected to achieve better decision performance.
- 3) We employ low-rank constraint to capture the intrinsic correlation across different views of features. Although the features from different views of lung CT images describe the characteristics of lungs and lesions from different aspects, the relationship structure among samples of different modalities has certain similarities. In our method, we capture the latent structural relationship of multi-view data by low-rank representation. In addition, although the data from different views are heterogeneous, we effectively fuse them through multi-kernel learning.
- 4) The experimental results on COVID-19 dataset show our method outperforms the state-of-the-art methods in terms of accuracy, sensitivity, specificity, and AUC. Fusing the labeled and unlabeled multi-view data by the proposed

method can significantly improve the diagnosis performance.

The rest of this paper is organized as follows. Related work is given in Section II. Our proposed approach is described in Section III. The experiments are presented in Section IV, with the results and discussion drawn in Section V. This paper is concluded in Section VI.

## II. RELATED WORK

### A. Multi-View Fusion

As a common space representation, canonical correlation analysis (CCA) is widely used in the field of multi-modal data fusion. CCA constructs the coordinated representation between two groups of random variables (views in our case) by calculating a linear projection that maximizes the correlation coefficient between them.

Suppose that there are two random variables  $\mathbf{X}_1 \in \mathbb{R}^{d_1}$  and  $\mathbf{X}_2 \in \mathbb{R}^{d_2}$ , with their covariances  $\Sigma_{11}$  and  $\Sigma_{22}$ . We use  $\Sigma_{12}$  to denote the cross-covariance of them. And CCA formula can be expressed as:

$$(\mathbf{P}_1^*, \mathbf{P}_2^*) = \min_{\mathbf{P}_1, \mathbf{P}_2} \frac{\mathbf{P}_1^T \Sigma_{12} \mathbf{P}_2}{\sqrt{\mathbf{P}_1^T \Sigma_{11} \mathbf{P}_1} \sqrt{\mathbf{P}_2^T \Sigma_{22} \mathbf{P}_2}} \quad (1)$$

where  $\mathbf{P}_1$  and  $\mathbf{P}_2$  are the corresponding projection matrices.

In previous work, CCA has proved to be a very effective unsupervised method of coordinated representation. For example, Klein et al. utilized CCA to perform cross-modal retrieval [26]. Sargin et al. used CCA to analyze the audio and video signal [27]. The original CCA can only calculate the linear projection between two groups of random variables, whereas the relationship between variables is usually nonlinear in many real-world problems. Kernel CCA (KCCA) introduces the idea of kernel function into CCA and uses kernel function to project low-dimensional data to feature space, which can be expressed as:

$$\begin{aligned} (\phi_1^*, \phi_2^*) &= \max_{\phi_1, \phi_2} \frac{\phi_1^T \mathbf{K}_1 \mathbf{K}_2 \phi_2}{\sqrt{\phi_1^T \mathbf{K}_1^2 \phi_1} \sqrt{\phi_2^T \mathbf{K}_2^2 \phi_2}} \\ \text{s.t. } \phi_1^T \mathbf{K}_1 \phi_1 &= \phi_2^T \mathbf{K}_2 \phi_2 = \mathbf{I} \end{aligned} \quad (2)$$

where  $\mathbf{X}_1$  and  $\mathbf{X}_2$  were respectively projected into feature space via  $\phi_1$  and  $\phi_2$ , respectively.  $\mathbf{K}_1$  and  $\mathbf{K}_2$  denote the corresponding kernel matrices.

However, KCCA is a nonparametric method, and the training cost is too high in dealing with large data sets. To solve these problems, Andrew et al. proposed deep CCA (DCCA) based on deep learning technique [28]. As an alternative to KCCA, DCCA solves the data scale issue, and it also brings a better coordinated representation attribute to the deep learning technique. DCCA constructs a deep neural network to maximize the correlation coefficient of the two random variables after projection, which is described as (3).

$$(\theta_1^*, \theta_2^*) = \max_{\theta_1, \theta_2} \text{corr}(f_1(\mathbf{X}_1; \theta_1), f_2(\mathbf{X}_2; \theta_2)) \quad (3)$$

Multi kernel learning (MKL) is a data fusion method based on support vector machines (SVM). Due to the benefit of the kernel function, the elements of the kernel function can be regarded as the inner product of the sample point in the feature space, and the MKL method can also well fuse the heterogeneous data. MKL has proven to be an effective model-based fusion approach. McFee et al. used MKL to fuse three different modal data from acoustic, semantic, and social views to perform similarity ranking on music artists [29]. Zhang et al. utilized three different modal data (MRI, PET, and CSF) to assist the diagnosis of Alzheimer's disease [30]. MKL is also a very popular method in fusing visual descriptors for object detection [31], multimedia event detection [32], multimodal sentiment analysis [33] and multimodal affect recognition [34].

Although MKL projects the original features into a feature space and increases the separability, the kernel matrix from different modalities is simply weighted sum without considering the structural information among different modal data.

### B. Semi-Supervised Learning

Although CT images are easily available, manually diagnosing such a larger number of CT images is time-consuming and error-prone. This situation leads to that a mass of CT images without diagnosis labels cannot be used in previous computer-aided diagnosis system. In clinical practice, it is much easier to acquire samples' features than their labels. Moreover, most of the human learning processes conducted under semi-supervised or unsupervised circumstances, and human can effectively utilize a large amount of unlabeled data to extract useful information for induction learning. Inspired by this phenomenon, we can make a reasonable assumption that the samples having similar data distribution usually belong to the same class, even if some samples are unlabeled. It is of great practical significance to develop a technique to make use of the features of these unlabeled samples.

Existing semi-supervised learning methods can be divided into five categories: generative model, self-training, co-training, semi-supervised support vector machine and graph-based methods [35]. Semi-supervised learning techniques in medical diagnosis aim to reduce the cost of obtaining training data by reducing labeling cost in the training process. For example, Jie et al. collected the MRI, FDG-PET and CSF data from Alzheimer's disease patients and healthy people and used group-sparse regularization and manifold regularization to improve the generalization performance of the model and maintain data distribution [36]. Peng et al. combined artificial immunity with semi-supervised learning and introduced Kent chaotic map into the algorithm for improving solution in the antibody cells feature vector space [37].

In particular, COVID-19, which has swept the world since the end of 2019, is extremely contagious and causes a rapid rise in cases worldwide. However, the fast speed of COVID-19 spreading leads to the shortage of medical resources and experts in the whole world, and a computer-aided diagnosis that can utilize both labeled and unlabeled samples has become a practical and urgent need now.

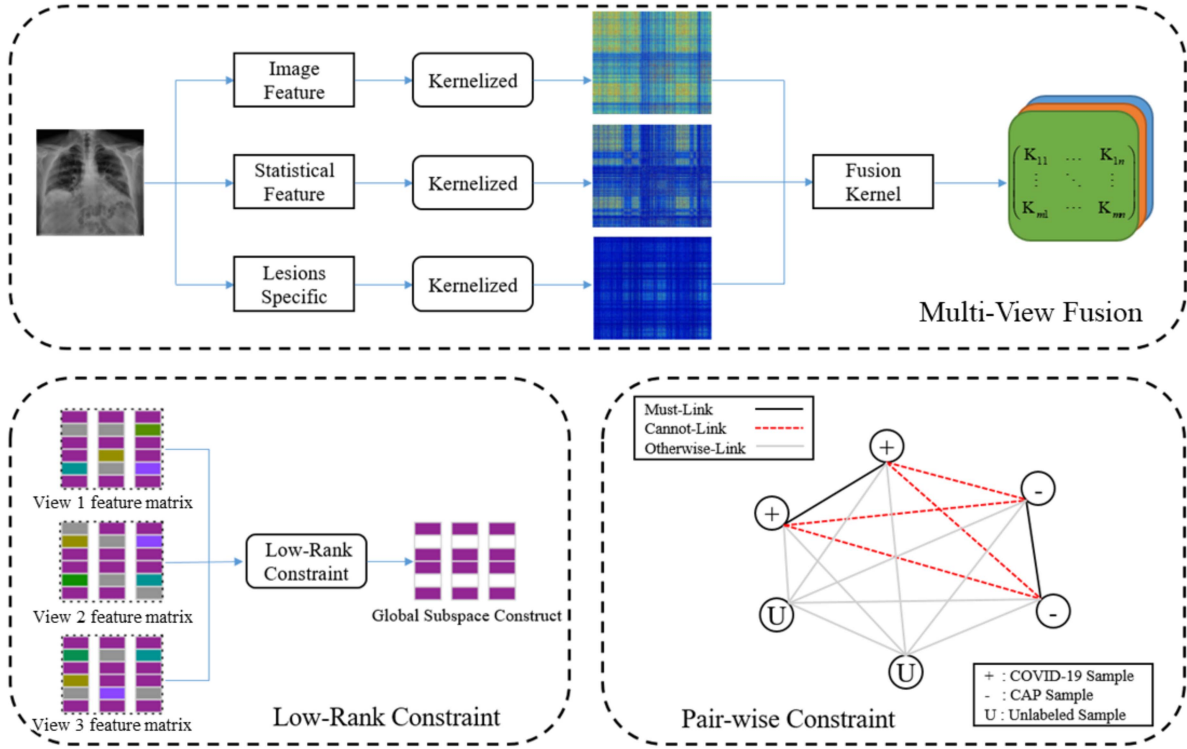


Fig. 2. The illustration of our proposed semi-supervised multi-view fusion based diagnosis model. Firstly, we extract three views of features from chest CT image, including image feature, statistical feature and lesions specific feature respectively. Then we apply low-rank constraint on coefficient matrix  $B$  in different views of features to explore the global subspace structure. Simultaneously, we impose the pair-wise constraint on the model to take advantage of unlabeled samples.

### III. SEMI-SUPERVISED MULTI-VIEW FUSION FOR COVID-19

#### A. Method Description

A large number of experiments and facts have proved that multi-view fusion can provide complementary information for the diagnosis of diseases [38], [39]. The different views of features we extracted from the CT images have discrimination for a certain extent, but simply concatenating them into a vector does not reflect the complementary characteristics of the information of different features. To fully explore the complementary information among different features, inspired by [30], we proposed a novel low-rank semi-supervised multi-view fusion diagnosis method. In our method, to align the feature dimension of each view features and improve the classification performance, we implicitly use feature space induced by kernel function  $\phi$ . Then, to fully utilize the unlabeled samples, we introduce a pairwise constraint to preserve the intrinsic structure of the data as well as the pair-wise constraints specified by users in the projected low-dimensional space. Simultaneously, the low-rank constraint is imposed to capture the intrinsic global subspace structure of the data. Fig. 2 demonstrates the framework of the proposed semi-supervised multi-view fusion for COVID-19.

#### B. Low-Rank Multi-View Fusion

Assume that there are  $m$  views of features, we denote them as  $\{\mathbf{X}^{(1)}, \mathbf{X}^{(2)}, \dots, \mathbf{X}^{(m)}\}$ .  $\mathbf{X}^{(i)} = [\mathbf{X}_1^{(i)}, \mathbf{X}_2^{(i)}, \dots, \mathbf{X}_N^{(i)}, \dots, \mathbf{X}_{N+M}^{(i)}]^T \in \mathbb{R}^{(N+M) \times d^{(m)}}$  represents the features of  $i_{th}$  view of  $N + M$  training samples whose first  $N$  samples are labeled and last  $M$  samples are unlabeled, and each view has same sample number, but feature dimension is not necessarily equal. The corresponding label for the first  $N$  samples can denote as  $\mathbf{Y} = [y_1, y_2, \dots, y_N] \in \mathbb{R}^N$ .

First, we consider a structural risk minimization (SRM) [40] function:

$$f = \min_{f \in F} \ell(f(\mathbf{x}), \mathbf{y}) + R(f) \quad (4)$$

In (4),  $\ell(\cdot)$  represents the sample loss on the labeled data, and  $R(\cdot)$  is a regularization term to limit the complexity of  $f$ . In this work, we adopt  $l_2$  square loss function. For now, we consider the single view features situation, and then (4) can be represented as:

$$\begin{aligned} f &= \min_{f \in F} \frac{1}{2} \sum_{i=1}^N (\mathbf{y}_i - f(\mathbf{x}_i))^2 + \frac{\lambda}{2} \|f\|^2 \\ &= \min_{\omega} \frac{1}{2} \sum_{i=1}^N (\mathbf{y}_i - \omega^T \mathbf{x}_i)^2 + \frac{\lambda}{2} \omega^T \omega \end{aligned} \quad (5)$$

Here, we use a feature map  $\phi(\cdot)$  to map sample feature to a reproducing kernel Hilbert space (RKHS). Eq. (5) can be rewritten as:

$$\min_{\omega} \frac{1}{2} \sum_{i=1}^N (\mathbf{y}_i - \omega^T \phi(\mathbf{x}_i))^2 + \frac{\lambda}{2} \omega^T \omega \quad (6)$$

We can easily get the solution of (6):

$$\omega = \frac{1}{\lambda} \sum_{i=1}^N (\mathbf{y}_i - \omega^T \phi(\mathbf{x}_i)) \phi(\mathbf{x}_i) = \Phi^T \beta \quad (7)$$

where  $\Phi = [\phi(\mathbf{x}_1), \phi(\mathbf{x}_2), \dots, \phi(\mathbf{x}_N)]^T$ . And  $\beta_i = \frac{1}{\lambda} (\mathbf{y}_i - \omega^T \phi(\mathbf{x}_i))$  is the  $i_{th}$  row of  $\beta$ . We use  $\Phi^T \beta$  to substitute the  $\omega$  in (6), then we can get:

$$\min_{\beta} \frac{1}{2} \beta^T \Phi \Phi^T \Phi \Phi^T \beta - \beta^T \Phi \Phi^T \mathbf{y} + \frac{1}{2} \mathbf{y}^T \mathbf{y} + \frac{\lambda}{2} \beta^T \Phi \Phi^T \beta \quad (8)$$

where  $\mathbf{y} = [y_1, y_2, \dots, y_N]^T$ . We notice that in (8), each  $\Phi$  followed with a  $\Phi^T$ . Then we can define a Gram matrix  $\mathbf{K} = \Phi \Phi^T$ , where  $\mathbf{K}_{i,j} = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j) = k(\mathbf{x}_i, \mathbf{x}_j)$ , ( $k(\cdot, \cdot)$  is the kernel function). In this paper, we use Gaussian kernel function, i.e., radial basis function (RBF), and it is defined as  $k(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}\right)$ . Then we can rewrite (8) as:

$$\min_{\beta} \frac{1}{2} \|\mathbf{y} - \mathbf{K}\beta\|_2^2 + \frac{\lambda}{2} \beta^T \mathbf{K} \beta \quad (9)$$

Eq. (9) is the general form of SRM in the RKHS, and then we can easily extent it to a multiple views form:

$$\min_{\beta} \sum_{j=1}^m \|\mathbf{y} - \mathbf{K}^{(j)} \beta^{(j)}\|_2^2 + \lambda \beta^{(j)T} \mathbf{K}^{(j)} \beta^{(j)} \quad (10)$$

In (10) we notice that the Gram matrix  $\mathbf{K}^{(j)}$  from different views have the same size, so the weight vectors  $\beta^{(j)}$  has the same length either. We define  $\mathbf{B} = [\beta^{(1)}, \beta^{(2)}, \dots, \beta^{(N)}]$ . Previous work [39] has proven that the low-rank constraint can effectively preserve the latent correlation information between different views. So, we adopt low-rank constraint on coefficient matrix  $\mathbf{B}$ :

$$\min_{\mathbf{B}} \sum_{j=1}^m \|\mathbf{y} - \mathbf{K}^{(j)} \beta^{(j)}\|_2^2 + \lambda \beta^{(j)T} \mathbf{K}^{(j)} \beta^{(j)} + \lambda_R \text{Rank}(\mathbf{B}) \quad (11)$$

### C. Pair-Wise Constraint

We notice that in (11)  $\mathbf{K}^{(j)} \in \mathbb{R}^{N \times N}$ , which means we construct Gram matrix only use the first  $N$  labeled data  $\{\mathbf{x}_1^{(m)}, \mathbf{x}_2^{(m)}, \dots, \mathbf{x}_N^{(m)}\}$ . Eq. (11) can extend to a semi-supervised form:

$$\min_{\mathbf{B}} \sum_{j=1}^m \|\mathbf{P}(\mathbf{y} - \mathbf{K}^{(j)} \beta^{(j)})\|_2^2 + \lambda \beta^{(j)T} \mathbf{K}^{(j)} \beta^{(j)} + \lambda_R \text{Rank}(\mathbf{B}) \quad (12)$$

where the Gram matrix  $\mathbf{K}$  in (12) is constructed by both labeled and unlabeled data  $\{\mathbf{x}_1^{(m)}, \mathbf{x}_2^{(m)}, \dots, \mathbf{x}_N^{(m)}, \dots, \mathbf{x}_{N+M}^{(m)}\}$ , and  $\mathbf{K}^{(j)} \in \mathbb{R}^{(N+M) \times (N+M)}$ . Meanwhile, we define a diagonal indicator matrix  $\mathbf{P} \in \mathbb{R}^{(N+M) \times (N+M)}$  with  $P_{i,i} = 1$  if  $\mathbf{x}_i^{(m)}$  is labeled data. Otherwise,  $P_{i,i} = 0$ .

Additionally, we add a pair-wise constraint regularization to further maintain the similar structure of nearest points. According to [41], we can impose pair-wise must-link ( $G_1$ ) and

cannot-link constraints ( $G_0$ ) on the model in (12). Here, we define a pair-wise affinity matrix as:

$$S_{i_1, i_2} = \begin{cases} -\left(\frac{1}{n^2} + \frac{\rho}{n_{G_1}}\right), & \text{if } (x_{i_1}, x_{i_2}) \in G_1 \\ -\left(\frac{1}{n^2} + \frac{\theta}{n_{G_0}}\right), & \text{if } (x_{i_1}, x_{i_2}) \in G_0 \\ -\frac{1}{n^2}, & \text{otherwise} \end{cases} \quad (13)$$

where  $n$  is the total sample number ( $n = N + M$ ),  $n_{G_1}$  and  $n_{G_0}$  are the number of cannot-link and must-link constraints, respectively, the parameter  $\rho$  and  $\theta$  are the scaling parameters to balance the contribution between must-linked constraints and cannot-link constraints, respectively. Empirically, we set  $\rho = 1$  and  $\theta > 1$ . By introducing another diagonal matrix  $\mathbf{D}$ , in which  $\mathbf{D}_{ii}$  value is the summation of  $i_{th}$  row in matrix  $\mathbf{S}$ , we can construct a Laplacian matrix  $\mathbf{L} = \mathbf{D} - \mathbf{S}$ , and our regularization term can be expressed as:

$$\begin{aligned} & \sum_{i_1, i_2=1}^{N+M} S_{i_1 i_2} (f(\mathbf{x}_{i_1}) - f(\mathbf{x}_{i_2}))^2 \\ &= \sum_{i_1, i_2=1}^{N+M} f(\mathbf{x}_{i_1}) \mathbf{L}_{i_1 i_2} f(\mathbf{x}_{i_2}) \\ &= \sum_{j=1}^m \beta^{(j)T} \mathbf{K}^{(j)} \mathbf{L} \mathbf{K}^{(j)} \beta^{(j)} \end{aligned} \quad (14)$$

According to (14) and (12), we can get our overall object function:

$$\min_{\mathbf{B}} \sum_{j=1}^m \|\mathbf{P}(\mathbf{y} - \mathbf{K}^{(j)} \beta^{(j)})\|_2^2 + \lambda \beta^{(j)T} \mathbf{K}^{(j)} \beta^{(j)} + \lambda_L \beta^{(j)T} \mathbf{K}^{(j)} \mathbf{L} \mathbf{K}^{(j)} \beta^{(j)} + \lambda_R \text{Rank}(\mathbf{B}) \quad (15)$$

### D. Optimization

To solve the object function (15), we notice that the solution of  $\text{Rank}(\cdot)$  is an NP-hard problem, so we replace it by nuclear norm function  $\|\cdot\|_*$ , which is the summation of the singular values. By introducing one auxiliary variable to make the problem separable, this problem can be solved in the alternating direction method of multipliers framework (ADMM). The object function can be rewritten as:

$$\begin{aligned} & \min_{\mathbf{B}} \sum_{j=1}^m \|\mathbf{P}(\mathbf{y} - \mathbf{K}^{(j)} \beta^{(j)})\|_2^2 + \lambda \beta^{(j)T} \mathbf{K}^{(j)} \beta^{(j)} \\ & \quad + \lambda_L \beta^{(j)T} \mathbf{K}^{(j)} \mathbf{L} \mathbf{K}^{(j)} \beta^{(j)} + \lambda_R \|\mathbf{Q}\|_* \\ & \text{s.t. } \mathbf{Q} = \mathbf{B} \end{aligned} \quad (16)$$

The corresponding augmented Lagrangian function is:

$$\begin{aligned} \mathcal{L}(\mathbf{B}, \mathbf{Q}, \mathbf{C}) = & \min_{\mathbf{B}, \mathbf{Q}, \mathbf{C}} \|\mathbf{P}(\mathbf{y} - \mathbf{K}^{(j)} \beta^{(j)})\|_2^2 \\ & + \lambda \beta^{(j)T} \mathbf{K}^{(j)} \beta^{(j)} \\ & + \lambda_L \beta^{(j)T} \mathbf{K}^{(j)} \mathbf{L} \mathbf{K}^{(j)} \beta^{(j)} + \lambda_R \|\mathbf{Q}\|_* \\ & + \langle \mathbf{C}, \mathbf{B} - \mathbf{Q} \rangle \\ & + \frac{\lambda_C}{2} \|\mathbf{B} - \mathbf{Q}\|_F^2 \end{aligned} \quad (17)$$

where  $\langle X, Y \rangle = \text{tr}(X^T Y)$ .  $\lambda_C > 0$  is a penalty parameter and  $C$  is the Lagrangian multiplier. According to the ADMM, we update one variable and fix the others in each iteration. We conduct our optimization progress as follows.

**Updating  $B$ :** According to the  $m$  views of features, we divide the problem of updating  $B$  into  $m$  sub-problems of updating  $\beta^{(j)}$ . Fix the other variables and update  $B$  by solving the following problem:

$$\begin{aligned} \min_{\beta} & \|P(y - K^{(j)}\beta^{(j)})\|_2^2 + \lambda\beta^{(j)T} K^{(j)}\beta^{(j)} \\ & + \lambda_L \beta^{(j)T} K^{(j)} L K^{(j)} \beta^{(j)} \\ & + \langle c^{(j)}, \beta^{(j)} - q^{(j)} \rangle + \frac{\lambda_C}{2} \|\beta^{(j)} - q^{(j)}\|_2^2 \end{aligned} \quad (18)$$

where  $\beta^{(j)}$ ,  $q^{(j)}$  and  $c^{(j)}$  denote the  $j_{th}$  column of the original matrix. Then we set the derivative of  $\beta^{(j)}$  to be zero, we obtain the solution:

$$\begin{aligned} \beta^{(j)} &= \left( 2K^{(j)} P K^{(j)} + 2\lambda K^{(j)} + 2\lambda_L K^{(j)} L K^{(j)} + \rho I \right)^{-1} \\ &\cdot \left( 2K^{(j)} P Y - c^{(j)} + \rho q^{(j)} \right) \end{aligned} \quad (19)$$

**Updating  $Q$ :** We fix the other variables and update  $Q$  by solving the following problem:

$$\min_Q \lambda_R \|Q\|_* + \langle C, B - Q \rangle + \frac{\lambda_C}{2} \|B - Q\|_F^2 \quad (20)$$

We use the singular value thresholding (SVT) [42], [43] operator to get the closed-form solution of this problem:

$$Q = \mathcal{T}_{\frac{\lambda_R}{\lambda_C}} \left( B + \frac{C}{\lambda_C} \right) = U \mathcal{S}_{\frac{\lambda_R}{\lambda_C}}(\Sigma) V^T \quad (21)$$

where  $U \Sigma V^T$  is the result of singular value decomposition (SVD) of  $B + \frac{C}{\lambda_C}$ , and  $\mathcal{S}_{\frac{\lambda_R}{\lambda_C}}(\cdot)$  is the soft-thresholding [43] operator, which is defined as:

$$\mathcal{S}_{\theta}(x) = \begin{cases} x - \theta, & \text{if } x > \theta \\ x + \theta, & \text{if } x < -\theta \\ 0, & \text{otherwise} \end{cases} \quad (22)$$

In each iteration progress, after updating the variable  $B$  and  $Q$ , we will update the Lagrangian multiplier  $C$  for a faster convergence. The Lagrangian multiplier  $C$  can be updated as:

$$C \leftarrow C + \lambda_C (B - Q) \quad (23)$$

The solution algorithm of the proposed semi-supervised data fusion method is presented in Algorithm 1.

According to Algorithm 1, we have solved (15) and obtained the regression coefficient matrix  $B$ . According to the representer theorem [44], the kernelized regression classification function has a general form:  $f^{(j)}(x) = \sum_{i=1}^{N+M} \beta_i^{(j)T} k(x_i, x)$ , where  $\beta^{(j)} = (\beta_1^{(j)}, \beta_2^{(j)}, \dots, \beta_{N+M}^{(j)})^T$  is the  $j_{th}$  column of regression coefficient matrix  $B$ . In the problem of COVID-19 classification, we extend the above algorithm to multi-view fusion version. Here, we define our fused classifier as  $f(x) = \sum_{j=1}^m \sum_{i=1}^{N+M} \alpha_j \beta_i^{(j)} k(x_i, x)$ . In this paper, to maintain the invariance of data scale, we adopt a constraint  $\sum_{j=1}^m \alpha_j = 1$

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**Algorithm 1:** Semi-Supervised Multi-View Data Fusion for Diagnosis of COVID-19.

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**Input:** Training labeled data  $\{x_1^{(i)}, x_2^{(i)}, \dots, x_N^{(i)}\}_{i=1}^I$ , corresponding label  $\{y_1, y_2, \dots, y_N\}$ , training unlabeled data  $\{x_{N+1}^{(i)}, x_{N+2}^{(i)}, \dots, x_{N+M}^{(i)}\}_{i=1}^I$ . Parameters  $\lambda, \lambda_L, \lambda_R$ . Laplacian matrix  $L$ .

**Output:** Matrix  $B$

**Initialization:**  $Q = 0, C = 0, \text{iter} = 0$

- (1) **While** (iter < Max iteration time) **do**
  - (2)   **For**  $j = \{1 \dots m\}$  **do**
  - (3)     Update  $\beta^{(j)}$  by using (19)
  - (4)   **End for**
  - (5)   Update  $Q$  by using (21)
  - (6)   Update  $C$  by using (23)
  - (7)   Iter+1
  - (8) **End while**
- 

on the coefficients of the different views of features, and we use grid search to find the best optimal combination by step 0.1.

#### IV. EXPERIMENTS

In this section, we will first detail our experiment setting. Then we evaluate our method performance through different extensive experiments by comparing with some state-of-art COVID-19 diagnosis methods and some widely used classification methods. Additionally, we analyze the influence of parameters and convergence of our method.

##### A. Dataset

In this paper, we utilized a dataset in our previous work [45]. Briefly, a total of 2522 patients were involved, in which 1495 COVID-19 cases and 1027 CAP cases. Of all the patients, 1258 are male and 1264 are female. Table I shows the age distribution. It could be seen that male CAP patients and female CAP patients share the similar age distribution, where patients aged between 50 and 70 take the highest ratio. But in COVID-19 patients, male patients concentrate in 30 to 40 and 50 to 60 years, and female patients concentrate in 50 to 70 years, in which female patients share the similar distribution in both two views of diseases. Details could be found in [45].

All the patients acquired chest CT scans with thin section. The CT scanner model used in the above hospital include: uCT 780 (UIH), Aquilion ONE (Toshiba), SOMATOM Force (Siemens), Optima CT520 (GE), Discovery CT750 (GE), LightSpeed 16 (GE), and SCENARIA (Hitachi). Each scan follows the same protocol, including: 120 kV, reconstructed slice thickness ranging from 0.625 to 2 mm, with breath hold at full inspiration.

##### B. Image Processing

In this paper, we preprocessed all CT images to segment the lung, lung lobes, and pulmonary out of the images according to a pretrained in-house research portal software [16] based on V-Net [46]. The details of the segmentation setting can be found

TABLE I  
AGE AND GENDER DISTRIBUTION OF THE DATASET

|                              | [10,20) | [20,30) | [30,40) | [40,50) | [50,60) | [60,70) | [70,80) | [80,90) | [90,100] |
|------------------------------|---------|---------|---------|---------|---------|---------|---------|---------|----------|
| <b>CAP(Female/Male)</b>      | 3/15    | 9/38    | 51/51   | 74/78   | 146/131 | 152/118 | 64/33   | 35/23   | 3/11     |
| <b>COVID-19(Female/Male)</b> | 5/11    | 50/83   | 132/200 | 144/147 | 204/167 | 155/121 | 31/34   | 3/6     | 0/1      |

TABLE II  
FEATURE DESCRIPTION OF THE COVID-19 DATASET

| View groups                  | Feature groups         | Dimension |
|------------------------------|------------------------|-----------|
| <b>Image view</b>            | Gray features          | 19        |
|                              | Texture features       | 74        |
| <b>Statistical view</b>      | Histogram features     | 30        |
|                              | Intensity features     | 2         |
|                              | Surface features       | 7         |
|                              | Age and Gender         | 2         |
| <b>Lesions specific view</b> | Number features        | 24        |
|                              | Volume features        | 33        |
|                              | Infected area features | 48        |

in [16]. Considering the intrinsic characteristics of each feature, we group them into three different views of features, including image features, statistical features, and lesions specific features, respectively, shown in Table II. Among all the nine feature groups in Table II, there are four groups of manual features extracted from the segmentation results. (1) Volume features: We use the infected areas of the whole lung, each lobe and pulmonary to calculate the whole lung infected ratio and each segment infected ratio by using lung field masks. (2) Infected lesion number: we count the infected areas of lungs, lung lobes, and segments, respectively. (3) Histogram distribution: We set the lung region image intensity value between -1350 and 150, and split them into 30 equal bins. Then the histogram distribution can be calculated by counting the frequency of intensity level in the infected region at each bin. (4) Surface area: Surface area features include the infection surface, the lung boundary surface, and the distance of each infection surface vertex to the nearest lung boundary surface. We put them into 4 groups, as 6, 9, 12, and 15 voxels (voxel spacing is 1.5mm).

### C. Experiments Setting

We utilize five evaluation metrics: ACC, SEN, SPE, the area under the receiver operating characteristic curve (AUC) and running time. Here, the ACC, SEN and SPE are defined as:  $ACC = \frac{TP+TN}{TP+TN+FP+FN}$ ,  $SPE = \frac{TN}{TN+FP}$ ,  $SEN = \frac{TP}{TP+FN}$  respectively, where TP/TN is the number of positive/negative samples correctly predicted and FP/FN is the number of positive/negative samples wrongly classified. For a medical diagnosis model, using only accuracy cannot evaluate a model comprehensively. Sensitivity (SEN) can express the proportion of actual positives that are correctly identified, and specificity (SPE) measures the proportion of actual negatives that are correctly

identified, which are both significant in medical diagnosis. To evaluate the performance of our model, we will take 5-fold cross validation strategy. Firstly, we divide the whole dataset into five exclusive subsets, which have the same or similar size. In each validation, we take one subset as test set and the others as training set. Meanwhile, the training set are labeled data and the test set are unlabeled data. The discriminative information in labeled data and the correlation structure in unlabeled data are both learned in our model, which is a kind of transductive semi-supervised learning strategy [47] to improve the generalization ability of model. We report the mean of above four evaluation metrics in 5-fold cross validation as our results. In addition, we also utilize  $p$ -value to measure the statistically significant improvement between the proposed method and the comparison method.

We compared our method with several classification methods and semi-supervised methods. To further evaluate the effectiveness and superiority of this method, we also conduct three different kinds of experiments: (1) We compare the result of multiple views features and single view features to prove that our model can capture the complementary information as expected. (2) Change the model parameter values within a certain range to test the sensitivity of the model to different parameters. (3) We adjust the ratio of the training set and test set to verify the ability of our semi-supervised model to utilize the features of the unlabeled test set data and the stability of our model under different ratios of labeled data.

## V. RESULTS AND DISCUSSION

### A. Classification Performance of the Proposed Method

To fairly evaluate the classification performance of proposed method in identifying CAP vs. COVID-19, we compared the performance of our method with several classic methods with five-fold cross validation strategy. For all methods, our dataset contains all 239 features. Except for our method, the input data of one patient of other methods is a 239-dimension vector by concatenating three views of features. We select 5 kinds of classical classification methods as the comparison methods to evaluate our method, including K-nearest neighbor (KNN), logistic regression (LR), support vector machine (SVM), random forests (RF), and neural networks (NN). In the KNN algorithm, we set the number of nearest neighbors to 5 and adopt Euclidian distance as the sample distance metric. In the SVM algorithm, we take linear kernel as our kernel function, and we define squared hinge as its loss function. In neural networks (NN), because our dataset has fewer feature dimensions and the size of dataset is far less than the massive data required by deep learning, we set up a shallow neural network. We set the batch

TABLE III  
CLASSIFICATION PERFORMANCE OF PROPOSED METHOD AND OTHER METHODS ON CAP vs. COVID-19

| Method                         | ACC(%)       | SEN(%)       | SPE(%)       | AUC(%)       | <i>p</i> -value (ACC) | Time(s) |
|--------------------------------|--------------|--------------|--------------|--------------|-----------------------|---------|
| LR                             | 89.99        | 92.23        | 86.50        | 95.74        | 0.01026               | 3.93    |
| KNN                            | 85.11        | 91.61        | 75.55        | 91.37        | 0.00017               | 6.17    |
| SVM                            | 89.96        | 91.97        | 85.48        | 95.95        | 0.02354               | 31.85   |
| RF                             | 89.48        | 90.69        | 87.83        | 95.47        | 0.00344               | 4.64    |
| NN                             | 89.16        | 91.11        | 86.34        | 95.21        | 0.00606               | 6.17    |
| L. Sun <i>et al.</i> [45]      | 91.03        | 92.52        | 88.91        | 96.35        | 0.04798               | 234.79  |
| X. Ouyang <i>et al.</i> [48]   | 88.57        | 90.42        | 87.82        | 93.72        | 0.00071               | 132.65  |
| T. Javaheri <i>et al.</i> [49] | 89.46        | 91.40        | 86.34        | 94.33        | 0.00373               | 190.49  |
| S. Hu <i>et al.</i> [50]       | 89.82        | 90.32        | 86.61        | 93.23        | 0.00375               | 240.23  |
| F. Shi <i>et al.</i> [51]      | 87.81        | 89.21        | 81.92        | 93.35        | 0.00098               | 39.56   |
| Ours(supervised)               | 91.75        | 93.04        | 89.85        | <b>96.71</b> | -                     | 27.36   |
| Ours                           | <b>92.02</b> | <b>93.11</b> | <b>90.44</b> | 96.29        | -                     | 41.73   |

size as 64, learning rate as 0.02, and the number of epochs as 300. For the other not mentioned parameters, we all choose the default or the optimal parameter setting. All these mentioned classical classification methods are all supervised methods. We also selected several state-of-art COVID-19 diagnosis methods [45], [48], [49], [50], [51] as our comparison methods. All the selected state-of-art methods share the same classification task (COVID-19 vs. CAP) as our method. It was worth noting that CAP and COVID-19 share a certain degree of pathologic similarity, so Covid-19 vs. CAP was a more difficult classification task compared to COVID-19 vs. normal control. In order to compare the performance between methods fairly, we extend our proposed semi-supervised method into a supervised classification version. We list the performance of the above methods on COVID-19 vs. CAP in Table III, with the best of each evaluation metrics listed in bold. The results in Table III show that the performance of our method is superior to all comparison methods, and it achieves the accuracy of 92.02%, sensitivity of 93.11%, specificity of 90.44% and AUC of 0.9629. In addition, the supervised learning version of our method is also superior to all comparing supervised learning methods with the performance of accuracy of 91.75%, sensitivity of 93.04%, specificity of 89.85% and AUC of 0.9671.

In addition, we adopt the following semi-supervised learning classification methods for comparison, including TSVM [52], EAGR [53] and SMIR [54] and ASFS [21]. The methods mentioned above and our method belong to the transductive semi-supervised learning approach. TSVM first uses all labeled samples to train a classifier and then uses all the samples to iteratively retrain and adjust the classifier to obtain an optimal classification boundary. It exchanges labels at each step to improve the objective function. Efficient anchor graph regularization (EAGR) is a graph-based semi-supervised learning method based on anchor graph regularization (AGR) technique. AGR can explore the underlying structure of the whole dataset to build a regularization framework with both data points and anchors. By a fast local anchor embedding method and adjacency matrix among

anchors, EAGR improves the shortcomings in anchor graph construction and regularization. Squared-loss mutual information regularization (SMIR) based on the information maximization principle designates the squared-loss mutual information (SMI) [55] as the information measure to be maximized. SMIR was proposed by introducing an SMI approximator with no logarithm inside. Table IV shows the performance comparison of the above methods on COVID-19 vs. CAP, with the best of each evaluation metrics listed in bold. As can be seen from Table IV, our method achieves the best results in ACC, SEN, SPE and AUC.

Furthermore, *p*-value of ACC between our method and the comparison methods in Tables III and IV are less than 0.05, which means those improvements by our method are statistically significant. We also calculated the complexity of the proposed algorithm and compared the time cost of our proposed method and other methods. The total computational complexity of Algorithm 1 is  $O(Zm(N + M)^3 + Zm^2(N + M))$ , where *Z* indicates the max iteration time. As can be seen from Tables III and IV, our method achieves higher computational efficiency than the SOTA diagnosis methods [45], [48], [49], [50]. Although some classification methods, such as LR, KNN, RF, and NN, have shorter computation time, their accuracies are significantly lower than ours. From Table IV, we find our method outperforms other semi-supervised methods of TSVM, SMIR and ASFS in diagnosis performance and computational efficiency. Although the calculation time of our method is longer than EAGR, it achieves significantly better diagnosis performance.

### B. Comparisons With Single-View Models

To show the improvement of multi-view data fusion to diagnosis, we compare the performance of the proposed method and its supervised learning version with single-view features and multiple views features separately. The results in Table V show that the performance of multi-view fusion data is statistically significantly better than that of any single view features under four evaluation metrics (ACC, SEN, SPE, AUC), no matter the

TABLE IV  
CLASSIFICATION PERFORMANCE OF PROPOSED METHOD AND SEMI-SUPERVISED METHODS (TSVM, EAGR, SMIR AND ASFS) ON CAP vs. COVID-19

| Method | ACC(%)       | SEN(%)       | SPE(%)       | AUC(%)       | p-value (ACC) | Time(s) |
|--------|--------------|--------------|--------------|--------------|---------------|---------|
| TSVM   | 90.08        | 91.55        | 87.93        | 95.46        | 0.02814       | 482.68  |
| EAGR   | 85.40        | 88.68        | 80.68        | 91.53        | 0.00053       | 27.64   |
| SMIR   | 84.48        | 87.75        | 79.71        | 91.88        | 0.00005       | 962.71  |
| ASFS   | 90.55        | 91.35        | 89.22        | 95.52        | 0.01330       | 43.27   |
| Ours   | <b>92.02</b> | <b>93.11</b> | <b>90.44</b> | <b>96.29</b> | -             | 41.73   |

TABLE V  
CLASSIFICATION PERFORMANCE OF PROPOSED METHOD WITH FUSION DATA AND EACH SINGLE VIEW DATA ON CAP vs. COVID-19

|                      | Method                | ACC(%)       | SEN(%)       | SPE(%)       | AUC(%)       |
|----------------------|-----------------------|--------------|--------------|--------------|--------------|
| Ours                 | Image view            | 88.49        | 90.9         | 84.98        | 94.62        |
|                      | Statistical view      | 89.17        | 92.3         | 84.59        | 94.73        |
|                      | Lesions specific view | 78.72        | 79.18        | 78.05        | 86.14        |
|                      | Fusion data           | <b>92.02</b> | <b>93.11</b> | <b>90.44</b> | <b>96.29</b> |
| Ours<br>(supervised) | Image view            | 88.65        | 90.77        | 85.56        | 94.9         |
|                      | Statistical view      | 89.76        | 92.84        | 85.45        | 95.56        |
|                      | Lesions specific view | 81.94        | 85.34        | 76.98        | 88.96        |
|                      | Fusion data           | <b>91.75</b> | <b>93.04</b> | <b>89.85</b> | <b>96.71</b> |

proposed method or its supervised learning version, especially the accuracy is improved by 2.85–13.30% compared with single view features. It is obvious that the proposed method and its supervised learning version with the multi-view fusion data have better performance than those with single-view features. Our experiments also show that our proposed method can effectively capture useful latent information between different views and construct a robust classifier for recognizing multi-view data.

### C. Performance With Different Combination Schemes

To visually show the performance of our proposed method of different combination of three views of features from CT images, we use the grid search to set the parameter combinations. Firstly, we apply a constraint  $\sum_{j=1}^m \alpha_j = 1$  to ensure data scale consistency, and then we use grid search method to search the weight of each view from 0 to 1 with a step size of 0.1. Fig. 3 shows the accuracy, sensitivity, specificity, and AUC of our proposed method under different weight combinations for different views. As shown in Fig. 3, in each heatmap, the positions of the three corner points in the upper triangle on the heatmap (upper left corner, upper right corner, and lower left corner) respectively represent the performance of the corresponding single view feature data (lesions specific feature, statistical feature, image feature), while the three edges (top edge, left edge and diagonal edge) represent the performance of the combination of two views of feature data. We can obviously observe in the heatmap, our model performs better, when it fuses all views of features than using two views or single view in the classification of COVID-19 and CAP.

### D. Parameter Sensitiveness Analysis

In our objective function (15), we notice that there are three parameters, i.e.,  $\lambda$ ,  $\lambda_R$  and  $\lambda_L$ . In addition, we constructed the Gram matrix by adopting the radial basis function (RBF) to map the original data to the feature space. By kernel trick, the product of any two samples after mapping can be defined as:  $k(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}\right)$ , where  $\sigma$  is the width parameter of RBF. To summarize, we need to adjust the selection of four parameters in the process of optimizing the model, respectively: (1)  $\lambda$ , which controls the overall complexity of the model, makes the model have good generalization and prevents the overfitting; (2)  $\lambda_R$  adjusts the weight of low-rank term; (3) paired constraint term weights  $\lambda_L$ ; (4) width parameter in RBF kernel function  $\sigma$ . In our extensive experiments, we observed that when  $\lambda_L$  was set in a certain range [1, 100], the final accuracy of the model would not change by more than 2%, which indicated that our method was not sensitive to  $\lambda_L$ . We conduct experiments to test the influence of the remaining three ( $\lambda$ ,  $\lambda_R$  and  $\lambda_L$ ) parameters on the performance of our method, so as to verify the robustness of our model.

In this experiment, we conducted pair-wise combination of these three parameters and conducted three parametric sensitivity analysis experiments. In Fig. 4, when  $\sigma$  is in the range of [1, 10], the model performance does not change much. Our model performance will descend when  $\sigma$  value is too small ( $<2$ ). Since this parameter controls the element values of Gram matrix, too small parameter  $\sigma$  may reduce the vector separability after RBF kernel mapping. About the model parameter  $\lambda$ , we observed when the  $\lambda$  is too large or too small, the performance of the model will significantly decrease. This indicates that it is important to

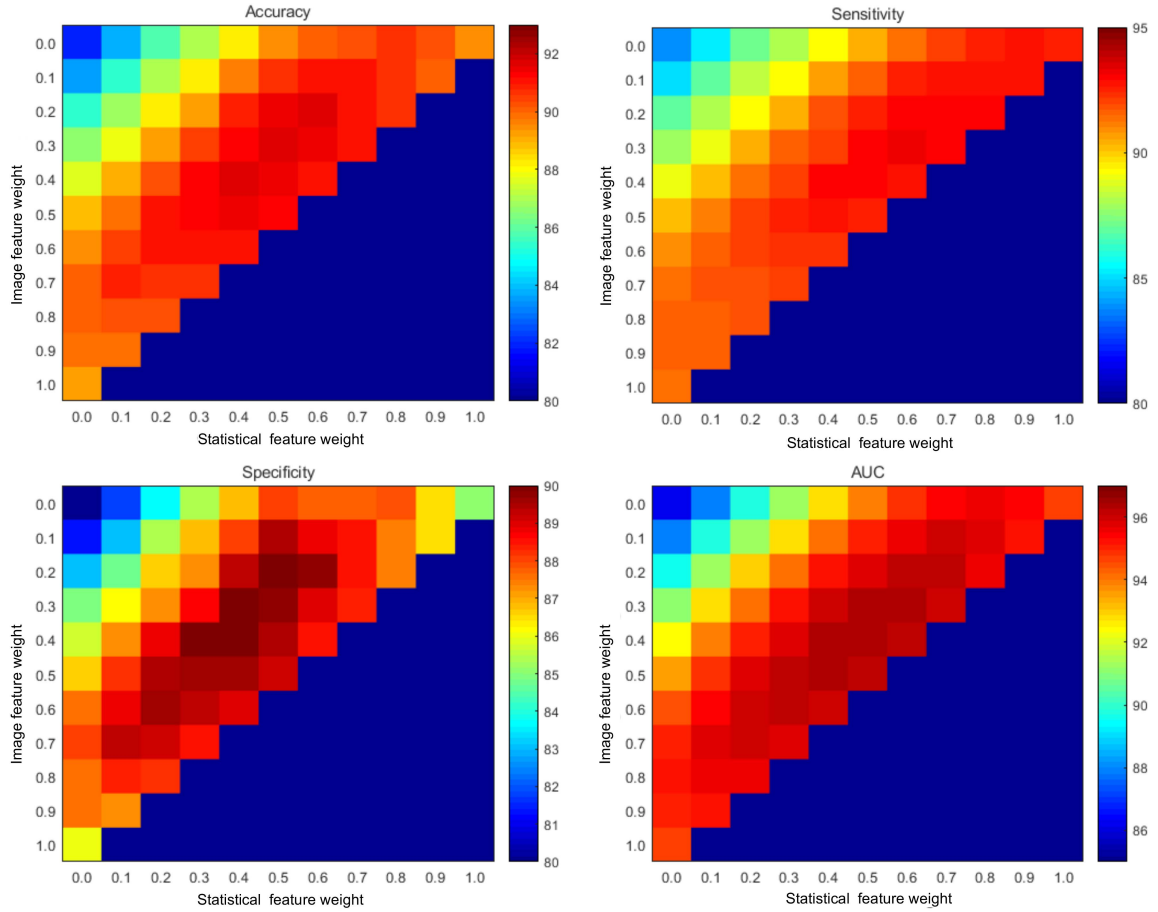


Fig. 3. The performance of proposed method with respect to different weight combinations of image feature, statistical feature and lesions specific feature.

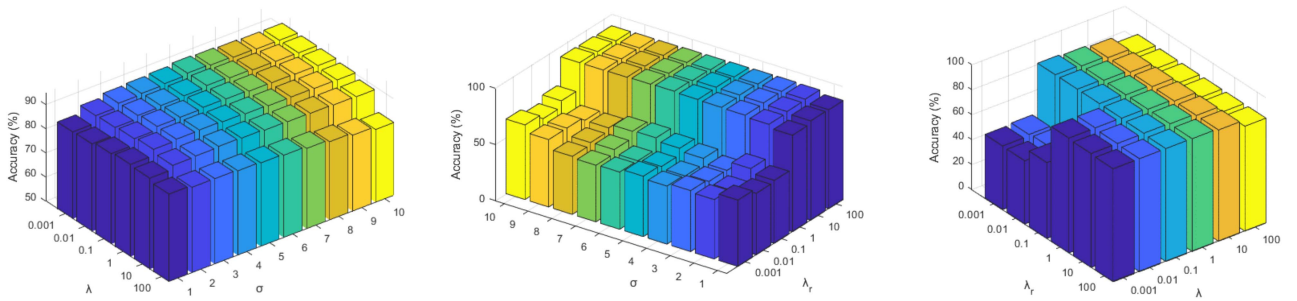


Fig. 4. The performance of proposed method vs. three parameters  $\lambda_R$ ,  $\lambda_L$  and  $\lambda$  on CAP VS. COVID-19.

choose the appropriate value for  $\lambda$  in our model. In addition, when the parameter  $\lambda_R$  is not too small ( $>1$ ), the performance of the model is promising, indicating that it is necessary to introduce low-rank constraints in the multi-view fusion problem. Therefore, in most cases, our model is very robust to parameter changes.

#### E. Stability of the Proposed Method

By adjusting the ratio of labeled data to unlabeled data, we verify the robustness of the proposed semi-supervised diagnosis

method. To ensure the fairness of the experiment, we repeated each experiment for many times under the same ratio of training set, and the ratio of the training set ranged from 30% to 90%. The samples of training set and the test set were randomly selected. Fig. 5 shows the changes of performance (ACC, SEN, SPE and AUC) of proposed method and several semi-supervised learning methods under different training set ratios. As can be seen from Fig. 5, the performance of our method at different training set ratios is higher than that of the comparison methods. Moreover, it was worth noting that the accuracy of our model did not change more than 3% as the ratio of labeled data changed from

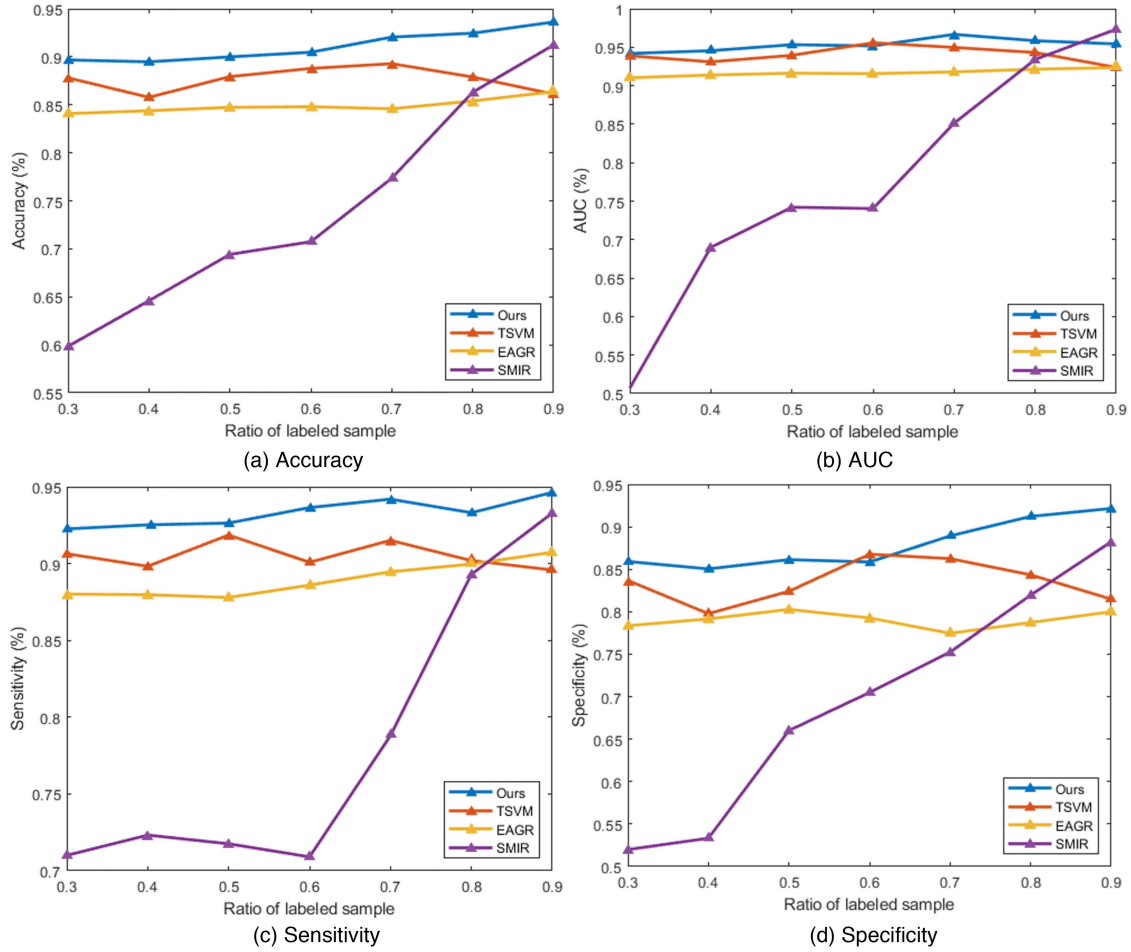


Fig. 5. The performance of proposed method and other semi-supervised methods under different labeled data ratios. (a) Accuracy. (b) AUC. (c) Sensitivity. (d) Specificity.

30% to 90%, indicating that our method could extract the most representative latent characteristics from a small amount of data. Even if the number of labeled samples was small, our model was also robust.

## VI. CONCLUSION

As the nucleic acid testing in large population reduce in many countries, it becomes increasingly urgent to develop accurate automatic diagnosis of COVID-19 using CT images. In this paper, we proposed a semi-supervised multi-view fusion analysis model for differential diagnosis of CAP and COVID-19 with chest CT images. Compared with the existing diagnostic models, the proposed model can effectively utilize a large number of unlabeled subjects. In addition, we introduce a low-rank constraint in multi-view fusion to capture the latent relationship among different views of features. Experimental results show that the performance of proposed method is superior to other methods in terms of accuracy, sensitivity, specificity and AUC. More importantly, the proposed semi-supervised multi-view fusion has a strong practical significance for computer aided diagnosis of COVID-19 with unlabeled subjects, especially in some countries without adequate medical resources and experts.

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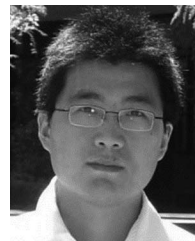
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